

# Karthik Jonnalagadda

NIIT University, Neemrana | [jonnalagadda.karthik22@st.niituniversity.in](mailto:jonnalagadda.karthik22@st.niituniversity.in) | LinkedIn: [jonnalagaddakarthik01](#) | Portfolio: [karthikjonnalagadda.github.io](#) | Ph: 9392991430

## Objective

*"Final-year Data Science engineering student with a solid academic foundation and hands-on experience in data analysis, machine learning, and software development; passionate about leveraging data-driven solutions to address real-world challenges and eager to contribute to impactful AI and data science projects."*

## Projects

### Multi-Modal Medical AI Assistant (RAG + Vision + OCR)

Aug 2025 – Dec 2025 | Team Size: 4

- **Technologies & Tools:** Python, PyTorch, Transformers, LangChain, LlamaIndex, sentence-transformers, DenseNet121, EasyOCR, Tesseract, FAISS, ChromaDB, BioGPT, spaCy, FastAPI, Streamlit, Docker, AWS
- Designed and implemented a production-grade multimodal healthcare AI system combining OCR, computer vision, and RAG to analyse lab reports, X-ray/MRI images, and patient symptoms.
- Built an OCR pipeline using EasyOCR and Tesseract to extract and normalise structured lab metrics (haemoglobin, WBC, platelets, etc.) from PDF and image-based reports, with automated reference-range flagging.
- Implemented a DenseNet121 / CheXNet vision model fine-tuned on the NIH Chest X-ray dataset (14-class), achieving structured radiological findings with per-class confidence scores.
- Engineered a **Retrieval-Augmented Generation (RAG)** pipeline using LangChain and ChromaDB/FAISS, grounding LLM responses (BioGPT) in verified PubMed, WHO, and ICD-10 medical knowledge — eliminating hallucinations.
- Integrated SHAP, LIME, and Grad-CAM explainability modules to provide transparent, evidence-based confidence scores for each diagnostic insight.
- Delivered a Streamlit clinical dashboard and FastAPI backend; deployed with Docker and docker-compose across three services (API · Dashboard · ChromaDB).

**Outcome:** System produces structured diagnostic outputs (possible conditions, confidence scores, supporting evidence, and cited medical references) with a full test suite covering OCR, vision, RAG, and API endpoints.

### SafeInbox: Real-Time Email Spam Detection

Feb 2025 – May 2025 | Team Size: 4

- **Technologies/Tools:** Python, Scikit-learn, React.js, Node.js, MongoDB, IMAP, Docker
- Built a machine learning pipeline (Naïve Bayes, Logistic Regression, SVM) for real-time spam detection.
- Integrated IMAP for Gmail fetching and developed a MERN-based dashboard with search, sort, and spam filters.
- Designed a secure frontend with authentication and MongoDB connectivity; deployed using Docker for scalability.
- Achieved **~94% spam detection accuracy** with reduced false positives.
- Tested email fetching pipeline, model outputs, and dashboard functionality; validated accuracy and UI behavior across scenarios.

### Empowering Rural Healthcare Through AI Solutions

Jan 2025 – Apr 2025 | Team Size: 4

- **Technologies/Tools:** AI/ML frameworks, Telemedicine platforms, Literature-based analysis
- Researched rural healthcare challenges (specialist shortage, delayed diagnostics, weak infrastructure).
- Evaluated AI solutions (telemedicine kiosks, AI diagnostics, maternal monitoring) for rural adaptability.
- Proposed an AI-driven framework for data collection, diagnostic support, patient triaging, and training.
- Case study showed **25% improvement in early disease detection** and **50% fewer unnecessary hospital visits**.

**Outcome:** Developed a scalable framework to improve accessibility, reduce delays, and optimize resources.

### Genomic Data Dimensionality Reduction & Clustering Analysis

Feb 2025 – Apr 2025 | Team Size: 2

- **Technologies/Tools:** Python, Scikit-learn, Pandas, Matplotlib, Seaborn, NumPy
- Applied SVD and NMF for dimensionality reduction on high-dimensional genomic datasets.
- Conducted exploratory data analysis (variable gene identification, correlation heatmaps, visualization of top genes).
- Performed K-Means clustering, achieving **ARI scores: 0.6592 (SVD) and 0.5611 (NMF)**.
- Validated biological separation using Random Forest classification on reduced features.

**Outcome:** Built a reproducible Python framework balancing SVD's numerical accuracy with NMF's interpretability.

## Education

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|---------------------------------------------------------------------|-------------|
| B-Tech (CSE-Data Science), NIIT University, Neemrana                | 2022 – 2026 |
| XII, Sri Gayatri Junior College, Hyderabad, Telangana (State Board) | 2022        |
| X, New Era School, Khammam, Telangana (State Board)                 | 2020        |

## Skills

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- **Languages:** Python, JavaScript, Java, R
- **Frontend:** HTML, CSS, Android Studio, React, Next.js
- **Backend:** Node.js, Express.js, Django, REST APIs
- **Databases:** MySQL, MongoDB
- **Machine Learning:** NumPy, Pandas, Matplotlib, Scikit-learn, TensorFlow, PyTorch, Transformers, sentence-transformers
- **AI / GenAI:** LangChain, LlamaIndex, RAG, ChromaDB, FAISS, BioGPT, EasyOCR, SHAP, LIME, Grad-CAM
- **Cloud Technologies:** Google Cloud, AWS (EC2, ECR), REST APIs
- **Frameworks & Tools:** Pinecone, Flask, FastAPI, Streamlit, Docker
- **Version Control:** Git, GitHub

## Certifications

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- Oracle Cloud Infrastructure 2025 Certified Data Science Professional – Oracle
- Introduction to Career Skills in Data Analytics – LinkedIn Learning